AQA 3.1.12 Acids and bases

AQA A-level Chemistry · Revision questions · Sheet 6 of 8

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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20 marks · about 25 minutes

Key information

- $pH = -\log_{10}[H^+]$ and $[H^+] = 10^{-pH}$
- Strong acid: $[H^+]$ = [acid]; for a diprotic strong acid $[H^+] = 2 \times$ [acid]
- $K_w = [H^+][OH^-]$ (1.00×10^{-14} mol^2 dm^{-6} at 25 °C); for a base $[H^+] = K_w \div [OH^-]$
- Weak acid on its own: $[H^+] = \sqrt{(K_a \times [HA])}$; $K_a = 10^{-pKa}$
- Buffer: $[H^+] = K_a \times [HA] \div [A^-]$ — use moles, the volume cancels

Section A: Short-answer questions

01 A strong acid H_2X dissociates in aqueous solution.

$H_2X(aq) \rightarrow 2H^+(aq) + X^{2-}(aq)$

What is the pH of a 0.020 mol dm^{-3} solution of this acid?

[1 mark]

A 1.00
B 1.40
C 1.70
D 2.00

02 Which can **not** function as a Brønsted–Lowry acid?

[1 mark]

A CH_3COO^-
B HCO_3^-
C H_3O^+
D NH_4^+

03 Calculate the volume of 0.100 mol dm^{-3} sodium hydroxide needed to reach the equivalence point with 25.0 cm^3 of 0.0800 mol dm^{-3} ethanoic acid.

[1 mark]

04 Methanoic acid has $pK_a = 3.75$. Calculate the value of K_a for methanoic acid.

[1 mark]

05 Calculate the pH of 0.075 mol dm^{-3} sodium hydroxide at 25 °C. Give your answer to 2 decimal places.
$K_w = 1.00 \times 10^{-14}$ mol^2 dm^{-6}

[2 marks]

Section B: Longer questions

06 At 40 °C the ionic product of water, $K_w = 2.92 \times 10^{-14}$ mol^2 dm^{-6}

06.1 Give the expression for K_w. Calculate the pH of pure water at 40 °C. Give your answer to 2 decimal places.

[3 marks]

06.2 35.0 cm^3 of 0.150 mol dm^{-3} aqueous sodium hydroxide are mixed with 20.0 cm^3 of a 0.100 mol dm^{-3} solution of hydrochloric acid. The temperature of the solution formed is 40 °C.

Calculate the pH of the solution formed. Give your answer to 2 decimal places.

[5 marks]

07 A buffer solution with a pH of 4.50 is made by dissolving *x* g of sodium propanoate (C_2H_5COONa) in a solution of propanoic acid. The final volume of buffer solution is 500 cm^3 and the final concentration of the propanoic acid is 0.250 mol dm^{-3}.

Calculate *x* in g. For propanoic acid, $K_a = 1.35 \times 10^{-5}$ mol dm^{-3}

[6 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	**B**	*Source: 7405/3 Jun 2022 Q14*	1
02	**A**	*Source: 7405/3 Jun 2022 Q13*	1
03	20.0 cm^3	*Source: Practice question (written for this set)*	1
04	$K_a = 10^{-3.75} = 1.78 \times 10^{-4}$ mol dm^{-3}	*Source: Practice question (written for this set)* Allow 2 sf or more	1
05	M1 $[H^+] = 1.00 \times 10^{-14} \div 0.075 = 1.33 \times 10^{-13}$ M2 pH = 12.88	*Source: Practice question (written for this set)*	2
06.1	M1 $K_w = [H^+][OH^-]$ M2 $[H^+] = \sqrt{(2.92 \times 10^{-14})}$ M3 pH = 6.77	*Source: 7405/1 Jun 2024 Q07*	3
06.2	M1 $n(OH^-) = 5.25 \times 10^{-3}$ **and** $n(H^+) = 2.00 \times 10^{-3}$ mol M2 excess $OH^- = 3.25 \times 10^{-3}$ mol M3 $[OH^-] = 3.25 \times 10^{-3} \div 55.0 \times 10^{-3} = 0.0591$ mol dm^{-3} M4 $[H^+] = 2.92 \times 10^{-14} \div 0.0591 = 4.94 \times 10^{-13}$ M5 pH = 12.31		5
07	M1 $[H^+] = 10^{-4.5} = 3.16 \times 10^{-5}$ mol dm^{-3} M2 $[C_2H_5COO^-] = [C_2H_5COOH] \times K_a \div [H^+] = 0.250 \times 1.35 \times 10^{-5} \div 3.16 \times 10^{-5}$ M3 $[C_2H_5COO^-] = 0.1068$ mol dm^{-3} M4 M_r sodium propanoate = 96(.0) M5 n(sodium propanoate) = 0.1068 × 0.5 = 0.0534 M6 mass = 0.0534 × 96 = 5.13 g	*Source: 7405/1 Nov 2020 Q04.4* M6 allow 5.09 to 5.14 g 10.25 g (not halved) scores 4 marks Henderson–Hasselbalch method allowed	6